order54 RESULTS
PREREG sha256 af2e5c10…1404 (PREREG.sha256), hashed before tests. Script sha in SHA256SUMS.
RG45c-1: PASS, Category A (source audit)
The written sources force the endpoint convention h(0) = 0. This is the midpoint convention, $\bar{B}1(0) = 0$. The sources are:
- order5 PREREG: "sgn 0 = 0", and E = erf(√π·) is odd.
- order5 zwegers.py uses np.sign.
- order18 a2.py defines B1b(0) = 0.
Nothing is invented. Explicit sketch: $A1(R_j) = \sqrt{(\pi/Q(u_j))}\cdot\Sigma_{r<P} m_r\cdot\bar{B}1((r+\theta_j)/P)$, with γ = ½ and EM jump −2 already folded in.
RG45c-2: FAIL as preregistered (numeric). The failure is a regime problem, not evidence against the formula.
- Wall 1 (u1, Q = 7/2):
  - Every endpoint case ($\theta+r \in Z$, $m_r \neq 0$; 222 cases in total) matches to ≤ 7.7e−16.
  - The one-sided control is off by up to 0.40, so it fails as the PREREG requires.
  - The non-endpoint cases miss by up to 0.19 (relative).
- Wall 2 (u2, Q = 595/2): the largest miss is 1.39. Only 18 of 630 cases pass.
- Overall: worst relative error 1.39 against the 1e−6 threshold. Verdict: FAIL. data/rg45c.json, logs/rg45c.log.
- POST-HOC diagnosis (logs/posthoc_diag.log; not a verdict):
  - The preregistered t-grid {4,2,1}e−3 is not asymptotic. The Poisson remainder along a line is about $\exp(-\pi^2/(t\cdot Q(u)\cdot P^2))$, and P reaches 28 or more. With wall 1 at t = 1e−3 that is about e^−3.6.
  - Worst wall-1 case: at t = 1e−4 the mismatch drops to 3.7e−16, matching the formula exactly.
  - Worst wall-2 case: the mismatch goes 0.97 → 0.95 → 0.18 at t = 1e−3, 1e−4, 2e−5. The wall-2 condition is $t \ll \pi^2/(297.5\cdot 784) \approx 4e{-}5$.
  - So both the formula and the h(0) = 0 endpoint are supported. A rerun needs a new PREREG with t scaled per case to the Poisson scale.
RG46: SKIPPED. A1(R_j) is a t^{−1/2} coefficient. It is not an explicit Ξ_compl or A1−Ξ_compl object, and nothing links it to the v12 shadow Ξ.
Ξ: unidentified. Bridge: NO.
Phase-aware k≠0 residual bound: not written; left Open.
Open next:
(a) order55: preregister a rerun of RG45c-2 with $t_k = c_k\cdot\pi^2/(40\cdot Q(u_j)\cdot P_j^2)$. Wall 2 needs about 1e5 lines × 500 terms, so use a reduced, preregistered sample and/or apply Poisson analytically along the lines.
(b) Then try matching Σ_j A1(R_j) to the v12 Ξ, and run RG46 only if an explicit object results.
(c) The k≠0 bound.
order54 NOTES
- Lock: CORE_FREEZE.
  - New files only, all in order54/.
  - order2/5/18/53 were read-only. order18 a2.py lines 1–30 were exec'd, not modified.
  - MAPS_CANDIDATES was not touched.
  - No Ξ/Π/α1 was invented. α is the existing companion shift.
  - RG* labels only. 35-component law vector. No push and no sends.
- Deviation: the first run finished all cases but crashed on JSON serialization of a numpy bool. I fixed only the serialization (bool casts), updated SHA256SUMS, and reran. The results were identical (worst relative error 1.386).
- posthoc_diag.py is explicitly post hoc and does not change the verdict.
order54 PREREG (Bridge C, 35-component law vector; RG* labels; CORE_FREEZE; new files only)
RG45c-1 (source audit, Category A): endpoint convention h(0)
Written sources: order5 PREREG §1.1 "sgn 0 = 0"; §1.3 E(z)=erf(√π z) (odd, E(0)=0); order5 zwegers.py np.sign (sign 0 = 0); order18 a2.py B1b(u)=0 at u∈Z. Hence h(0) = (E−sgn)(0)·G(0) = 0 = ½(h(0+)+h(0−)) (h odd): the midpoint convention, $\bar{B}1(0)=0$. Forced, not chosen.
RG45c-2 (numeric, Category A-numeric): A1(R_j) sawtooth formula
$R_j := ½\,\Sigma_{v\in Z^2}\,(E-\mathrm{sgn})(z_j(v))\,f(v)\,e^{-tQ(v)}$, $f = e(B(v,\beta)+xQ(v))$ (lib2

phase_ints), ν=α+v, y=t/2π,
z_1 = √10·ν1·√y (wall u1=(0,1), λ=ν1, θ1=α1), z_2 = (ν1+21ν2)√(2y/17) (wall u2=(21,−1), λ=ν1+21ν2, θ2=α1+21α2).
Prediction: √t·R_j → A1_j = √(π/Q(u_j)) · Σ_{r=0}^{P−1} m_r B̄1((r+θ_j)/P), Q(u1)=7/2, Q(u2)=595/2,
m_r = period mean of f on the line through base v=(r,0) in direction u_j (order53 RG45b), P = ℓ-period of m (lcm of line periods, from a2.period along (1,0)), B̄1(0)=0.
(Derivation: Poisson along lines gives √(π/(tQ(u_j)))·m_ℓ·h(√t(ℓ+θ)); EM jump J=h(0+)−h(0−)=−2, ∫h=0, h' even ⇒ no t^0 jump term; error O(t).)
Direct: line-by-line exact summation (b-range to Gaussian tail < 1e−18, λ range |λ|√t ≤ 9·scale), t ∈ {4e−3, 2e−3, 1e−3}; A1_num = quadratic-in-t Richardson extrapolation of √t·R_j to t=0.
Sample: XS[:6] × all 105 companions × 2 walls (fixed now; runtime-limited, no post-hoc change).
PASS iff max |A1_num − A1_pred| ≤ 1e−6·max(1,|A1_pred|) over the sample, AND the gating control FAILS:
 control = one-sided convention B̄1(0)=−½ (i.e. h(0)=h(0+)), required to miss by >1e−4 on ≥1 case with θ_j+r∈Z and m_r≠0; if no such case exists in the sample, the control is reported VOID and the verdict states the endpoint is untested numerically.
RG46: runs only if an explicit Ξ_compl / A1−Ξ_compl tied to the v12 shadow Ξ results. A1(R_j) alone is not such an object (it is a t^{−1/2} coefficient, not a shadow); expected SKIP.
Phase-aware k≠0 bound: not written in sources; not attempted.